Emad Gohari

Senior Machine Learning Engineer · ML Systems, NLP, GenAI — Toronto, ON, Canada

Contact Info

Email: E.Gohari.B@gmail.com
GitHub: github.com/emadg
LinkedIn: linkedin.com/in/emad-gohari
Website: EmadGohari.com

Summary

Senior machine learning engineer with 8 years of experience delivering ML, NLP, and LLM systems from research to production. At Thomson Reuters Labs, built reproducible training pipelines and PII-isolated infrastructure, shipped production GenAI summarization and semantic search, and contributed agent configuration and evaluation tooling to a commercially launched AI product. Collaborates with researchers, domain experts, and product engineers.

Technical Skills

ML & Deep Learning: PyTorch, Hugging Face Transformers, scikit-learn, XGBoost, Transformer Fine-Tuning (T5), LSTM, Clustering Models, Model Training & Evaluation (F1 Score, ROC-AUC, MRR, MAP, Recall@k)

MLOps, Data & Cloud: AWS SageMaker (Studio, Pipelines), MLflow, Apache Spark (PySpark), Snowflake, AWS CloudFormation, Data Pipelines, DAG Orchestration, PII Handling

GenAI & NLP: LLM Applications, Agentic Frameworks (PydanticAI), Prompt & Context Engineering, LLM Evaluation (LLM-as-Judge, SME Review), GPT-4, Claude, Sentence Embeddings, Semantic Search, Text Classification & Tagging, Knowledge Graphs & Ontologies

Languages & Libraries: Python, Java, Scala, Bash, pandas, NumPy, spaCy

Experience

Senior Machine Learning Engineer II (Research) at Thomson Reuters

Jul 2021 - Present
Toronto, Canada

Built ML systems at TR Labs for CLEAR's KYC/AML screening and investigations, and Checkpoint tax research. Worked with researchers, domain experts, and product engineers on data infrastructure and customer-facing features.

Work Projects

• **Model Retraining & Experimentation Pipelines: CLEAR** — 2026 | Internal Tool

Designed, implemented, and tested an on-demand SageMaker retraining pipeline, turning researchers' feature-generation, training, and evaluation code into reusable DAG steps. Enabled parallel experiments tracked in MLflow, cutting researcher time per round by an estimated ~40%. Extended reusable Python/PySpark components; built Snowflake processing and training proofs of concept and assessed migration gaps.

Tech & Skills: Python, AWS SageMaker Pipelines, MLflow, PySpark, Snowflake, XGBoost

• **GenAI Report Summarization: CLEAR Business AI** — 2024 | Production

Built an LLM summarization workflow live in production processing hundreds of reports daily, with selective XML extraction and targeted per-section calls. Integrated Bing Search, LLM verification, and source links to reduce same-name entity mismatches. Refined extraction and prompts through SME reviews; compared GPT-4 and Claude Sonnet on quality, latency, and cost to inform model selection.

Tech & Skills: Python, GPT-4, Claude Sonnet, Bing Search API, XML Processing, LLM Evaluation

• **Entity Resolution Data Infrastructure: CLEAR KYC/AML** — 2022 - 2024 | Internal Tool

Built research infrastructure for an entity-resolution system spanning ~800M entities and billions of documents. Designed a PII-isolated SageMaker/CloudFormation environment with researcher-specific profiles; defined a versioned JSON schema reconciling two annotation sources with Spark/Python normalization, merging, and conflict checks; and automated manual runs into scheduled SageMaker Pipelines.

Tech & Skills: Python, Apache Spark (PySpark), AWS SageMaker, CloudFormation

• **Agent Configuration & Evaluation: CLEAR Investigate** — 2025 - 2026 | Production

Contributed to commercially launched CLEAR Investigate. Built a central PydanticAI layer for prompts, tool descriptions/availability, and agent/subagent wiring, enabling agent reconfiguration from one place. Migrated existing LLM-as-judge tooling into a repository for reuse and future CI/CD integration.

Tech & Skills: Python, PydanticAI, Claude, Agent Configuration, LLM-as-Judge

• **Semantic Search Query Understanding: Checkpoint** — 2021 | Production

Shipped a sentence-embedding query-intent classifier separating broad conceptual queries from specific tax lookups to promote higher-level documents. Improved access to conceptual documents by 95% within a ~100ms latency budget, addressing a retrieval gap identified with SMEs and customer feedback.

Tech & Skills: Python, Sentence Embeddings, Text Classification, Semantic Search

NLP & Machine Learning Engineer at INAGO Inc.

Jul 2018 - Jun 2021
Toronto, Canada

Built NLP systems across data preparation, training, evaluation, and application integration with research partners. Trained domain-specific Word2Vec embeddings and LSTM NLU models with interpretability testing.

Work Projects

• **Transformer Fine-Tuning for Question Generation** — 2020 | Research

Fine-tuned T5 in PyTorch/Hugging Face to generate questions from documents, cutting manual data-curation effort by 40%. Experimented with input representations and evaluated output using BLEURT with university research partners.

Tech & Skills: Python, PyTorch, Hugging Face Transformers, T5, BLEURT

• **Conversational Question Answering with NLP & LSTM Models** — 2018 | Research

Developed conversational question answering combining document analysis and LSTM-based text tagging. Used semi-automatically generated ontologies to interpret questions and intent; experimented with knowledge graphs in an industry research collaboration.

Tech & Skills: Python, spaCy, LSTM, Text Tagging, Ontologies, Knowledge Graphs

Education

M.Sc. in Computer Science, York University

2014 - 2018
Toronto, Canada

Thesis: Interactive Question Answering Using Frame-based Knowledge Representation

B.Sc. in Computer Engineering, Amirkabir University of Technology

2010 - 2014
Tehran, Iran

Publications

Question-worthy Sentence Selection for Question Generation

Canadian AI 2020, 2020 · co-authored

Time Aware Topic-based Recommender System

Big Data & Information Analytics, 2016 · co-authored

A Study on Prediction of User's Tendency Toward Purchases in Websites based on Behavior Models

Information and Knowledge Technology (IKT), 6th Conference, IEEE, 2014 · co-authored